MVP validation

Evidence draft, 2026-11-05. Human review, real-data numbers and public-app evidence are pending. This document accurately reports completed software/fixture checks; it is not a claim that the real-data MVP or validation is complete.

This app helps a CPSC epidemiology analyst decide which treadmill injury mechanisms among adults 65-120 deserve closer investigation. It combines estimated emergency-department (ED) visits, recorded severe outcomes, primary diagnoses and supporting narratives. The full 2025 source passed 410,201-row, 25-column and calendar-year checks and produced 188 screening candidates. No finalized injury cohort exists. Ryan Yu explicitly confirms personal review of the original 171 decisions and supplied all remaining inclusion adjudications. Human inclusion decisions total 109 Include / 79 Exclude. There are 184 complete review entries; four included cases still lack defensible mechanism approval because use context is unstated. These are review-progress counts, not released injury findings. The original source/AI fields, review template, prior approvals and policy fingerprint remain preserved.

1. Design

The CPSC NEISS FAQ describes surveillance estimates and follow-back investigation, and distinguishes product association from causation. This supports an investigation-screening use case. Our particular interface, role scenarios and prioritization workflow are design hypotheses; no interview, usability study or observed analyst adoption is claimed. PENDING: team/user corroboration of decision constraints, workflow and usefulness.

The intended decision is where to spend limited follow-up attention, rather than whether a product is defective. Burden and severe share answer different questions: estimated volume versus recorded outcomes among visits. Diagnosis and narratives add context; visible evidence and missing outcomes help prevent overreading a high percentage supported by few records. The following eight changes extend the professor's starting build and remain protected by automated tests.

Change request	Implemented behavior and decision rationale
CR-01	Selected severe burden alongside all-outcome severe share, retaining mechanism/search. Prevents a severe-only filter from flattening every nonempty percentage to 100%.
CR-02	Visible mechanism record counts and unknown-disposition counts/weights. Makes evidence volume and missingness inspectable without inventing precision thresholds.
CR-03	Primary-diagnosis baseline beside selected composition under severity filtering. Keeps clinical context from being silently redefined.
CR-04	Matching IDs and original narratives after all active filters. Enables users to examine what a literal search actually selected.
CR-05	Reset and session-specific Undo reset. Restores the complete prior mechanism/severity/search selection for comparison.
CR-06	Unknown share and none/all-unknown-severe endpoints. Shows the decision impact of missing outcomes without imputing values.

Change request	Implemented behavior and decision rationale
CR-07	In-app mechanism definitions, version and initiating-event precedence. Makes taxonomy interpretation visible.
CR-08	Full-precision contribution reconciliation and CSV with filter/period/data hash. Explains display rounding and supports independent audit.

The team deliberately did not build exposure-adjusted risk, geographic/brand attribution, prediction, causal defect detection, significance claims or automatic investigation recommendations. The data and design do not support them. Source period is fixed to 2025; year-range controls and trends would require additional agreed artifacts and sampling interpretation. Counts are not confidence intervals, and no design-based uncertainty is estimated.

Meaningful alternatives are the official NEISS query builder, an independent spreadsheet/script, and CPSC follow-back investigation. The official query provides selectable confidence intervals/CV and downloadable records; it does not automatically reproduce our narrative-screened, manually classified cohort. A separate analysis can explore broader fall-involving definitions but takes reconciliation work. Follow-back can supply circumstances absent from short narratives; this dashboard cannot replace it. A database or automated classifier would not resolve the present evidence/approval gap. Small, precomputed flat files keep interactions in memory and avoid exposing preparation inputs.

Preparation, loading, filtering, aggregation and UI wiring remain separate. Source and candidate fingerprints bind a completed review to its exact export; the loader rejects missing, edited, incomplete or sample artifacts instead of silently displaying national estimates. The injury-focused policy narrows the original broader involvement contract. Ryan explicitly accepts the matching policy and confirms his case review. The policy document retains its original authorship/fingerprint, and the actual acceptance is preserved in private review documentation. His subsequent version 1.1 taxonomy authorization supersedes only the old mechanism-version reference; inclusion scope and the original prompt are not silently changed. All records and taxonomy boundaries require real human verification; AI agreement does not satisfy that gate.

PENDING LIVE EVIDENCE: initial public screenshot with period, labels and selections; user/team assessment of whether these features support the intended decision. The code and tests are evidence of implemented behavior, not of actual user benefit.

2. Cases

These are illustrative role scenarios, not actual people or completed user research. The official surveillance/follow-back context supports the general tasks; individual priorities and decision constraints are assumptions the team must verify. For each case, capture the public app with active controls and record the actual interpretation, alternatives and limitations. No production result or screenshot is invented.

Scenario and decision context	Interaction and interpretation to verify
Surveillance analyst allocating follow-up attention across mechanisms	Start with all 2025 records. Compare burden and severe share, supporting counts and primary diagnoses. PENDING: actual ranked mechanism, weights, evidence counts and screenshot. More weighted ED visits supports a volume question, not higher risk per user-hour. Consider activity exposure, healthcare access and review exclusions before prioritizing.

Scenario and decision context	Interaction and interpretation to verify
Incident reviewer checking a suspected belt-involved severe-fall pattern	Select Fall, Severe and literal belt search. Read matching narratives and compare with the same-search all-outcome baseline. PENDING: actual membership, values, narrative interpretation and screenshot. Belt may be a landing surface; a match does not prove belt initiation, defect or causation. An empty result is a valid result.
Evidence-quality lead deciding whether the severe share supports a defensible comparison	Inspect record count, unknown outcomes and conditional endpoints; compare primary-diagnosis mix. PENDING: actual unknown weight/share, endpoints and screenshot. Consider non-ED injuries left out, small supporting samples and missing clinical context. The endpoints are conditional assumptions, not probability intervals.
Analyst comparing a narrow subset with the starting population	Exercise Reset, Undo and a no-match phrase. PENDING: live controls, restored selection and screenshots. Confirm that comparisons retain the intended population and that empty results show unavailable percentages rather than misleading zeros.

Ryan's claimed individual HW11 case is preserved in cases/ryan-treadmill-falls.md; the cleaner memo is cases/ryan-treadmill-falls-final.md, and its source/claim audit is cases/ryan-treadmill-falls-evidence.md. Alex, a fictional CPSC epidemiology analyst, must decide whether older-adult treadmill falls deserve priority. The planned public selections are: all mechanisms/outcomes and blank search; Fall with all outcomes; then Fall + Severe + literal belt. No live observations or personal verdict are supplied. Codebook 1.1 classifies initiating mechanisms, so entering/exiting and belt-initiated falls can lie outside Fall. Alex must decide whether that definition answers the priority question before relying on a rank. Neither the separate HW11 PDF nor a teammate's verdict was fabricated.

The CPSC estimates explanation describes unequal weights and survey uncertainty. Sample counts do not replace design-based uncertainty or an exposure denominator. An apparent severe-share difference would not by itself settle investigation priority, especially when missing outcomes, diagnostic mix or medical triggers differ. The team must support each final case conclusion using actual observations and external context. No actual user interview or completed teammate case was found in the repository.

Scope-sensitive review cases are handled offline before these uses: medical-only visits versus medically triggered injury, mixed workouts versus unlinked history, direct equipment/cord contact versus unrelated hazards, and injury evidence versus isolated pain. Ryan resolved the eight inclusion decisions: seven Exclude, one Include with Unknown/Unclassified. The version 1.1 extension adds Non-use equipment/cord-related falls and explicitly covers an unpowered supporting-surface Fall. Four conditional non-use mappings remain held because the source does not establish use context. Their inclusion is approved. Do not infer non-use from omitted detail or turn a known uncovered event into Unknown to pass software. All 188 narratives were reassessed; no definite additional prior-label change was found among the 171 existing approvals, which remain unchanged.

PENDING CASE CONCLUSIONS: after genuine review and deployment, state for each scenario whether the analyst is better off, which verified observation supports that conclusion, what outside evidence is still needed and when the app should not be used. Until then, usefulness is plausible rather than demonstrated. This prevents software tests from being described as user validation.

3. Numbers

One row represents one ED visit. Let T be selected survey-weight sum; S the weight for dispositions 2/4/5/8; U the weight for disposition 9, missing or unrecognized. Codes 0/1/6 are other known outcomes. We retain unknown weight in the severe-share denominator, do not rescale weights after exclusion, and use primary Diagnosis only. These are team conventions, not a clinical severity scale.

Metric	Correct definition and independent check
Included records and weighted ED visits	Unique selected visit count and T. Compare independent ID membership/count and Decimal weight sum with actual filters/headline.
Severe count and weighted severe share	Count of disposition 2/4/5/8 records and 100*S/T. Verify severe-filter membership and weighted sums separately. Empty denominator is unavailable.
Mechanism totals	Sum original selected weights within each reviewed label; contributions must reconcile with T. Exclusions and unresolved records never enter the cohort.
Diagnosis composition	Primary-diagnosis weight divided by that mechanism's selected weight. Preserve missing and unmapped buckets; compare bucket sets/weights/shares and no duplicate contributions.
Unknown outcomes and sensitivity	Count and U; 100*U/T; conditional endpoints 100*S/T and 100*(S+U)/T. No missing value is recoded or imputed.
Baselines and exported precision	Baseline retains mechanism/search and removes severity only. Compare baseline severe share and selected severe burden independently. Downloaded CSV contributions must match independent unrounded numbers; final Weight must equal its original candidate value as Decimal.

validation/independent_numbers.py uses csv, Decimal and loops without importing dashboard aggregations. validation/verify_dashboard.py extends the independent checks, then compares them with actual production filter/view builders and reconciliation exports. It covers all records, each available mechanism, Severe/Other known/Unknown, literal belt/metacharacter searches, combined Fall/Severe/belt and empty selections. Counts, IDs and bucket sets are exact; numerical comparison tolerance is max(1e-8 absolute, 1e-12 relative). Native public-browser presentation remains a separate check.

The reproducible fixture report is validation/results/dashboard_fixture_results.json. Records A-E and the historical toy label Contact are INVENTED, not production NEISS categories/findings. Actual fixture results and hand arithmetic agree:

Invented fixture check	Independently verified result
Count and total	5 records; 10+30+20+40+20 = 120 weight
Recorded severe	3 records; 10+20+20 = 50 weight; 100*50/120 = 41.666666...%
Mechanism weights	Fall 40; toy Contact 60; Unknown/Unclassified 20
Primary diagnosis within Fall	57: 10/40 = 25%; 53: 30/40 = 75%
Primary diagnosis within toy Contact	57: 20/60 = 33.333333...%; missing: 40/60 = 66.666666...%
Unknown outcomes	1 record; 40 weight; 100*40/120 = 33.333333...%
Conditional endpoints	41.666666...% if none unknown severe; 100*(50+40)/120 = 75% if all unknown severe
Display rounding	Three 10.4 contributions sum to 31.2. Rounded headline 31 can differ from separately rounded contributions summing to 30; exported contributions remain unrounded.

During implementation the extended checker caught noncanonical diagnosis strings (53.0 versus 53) and incorrectly typed string weights in the new fixture runner. Canonical integer-code parsing and production-type normalization fixed the validation inputs; no metric definition or application aggregation was changed to force agreement. A regression test also injects an incorrect export value and confirms rejection. No unexplained fixture discrepancy remains.

PENDING REAL DATA: retained count, T, severe count/S/T, mechanism/diagnosis totals, U and both endpoints, final SHA-256, actual public outputs and CSV precision. No actual production cohort exists because four mechanism approvals remain unresolved, even though all inclusion decisions and the policy acceptance are recorded. Source/candidate/policy fingerprints match their preserved evidence; this verifies file identity, not human judgment. Candidate/AI counts and annual-source totals do not substitute for this evidence. The guarded finalization workflow will run these checks automatically after actual review; validation/README.md gives the separate executable command. The team must verify and defend the definitions and resulting interpretations.

4. Tests

The full actual repository suite passed **178 tests**, **0 failures**, **0 errors** and **0 skipped** on Windows/Python 3.13.15 on 2026-11-05; JUnit elapsed time 37.128 seconds. Fifteen new tests beyond the prior 163 protect the revised taxonomy (nine), video input/evidence checks (five), and exact Git artifact bytes (one). Command: python -B -m pytest -q -p no:cacheprovider, with JUnit and temporary files outside the repository. Results and module counts are in validation/results/test_summary.json. No success is inferred from an earlier run.

Existing tests cover preparation eligibility and annual source baselines, preserved fields/fingerprints, incomplete-review rejection, loader/provenance failures, empty/sample behavior, all five views, filters, cache invalidation and session isolation. tests/test_mvp_changes.py protects all eight change requests; tests/test_independent_numbers.py compares independent arithmetic with chart/table data. Original request descriptions and implementation commits in change-requests.md are preserved.

The review-transfer tests reject absent/deferred approvals, missing reviewer/explicit human verification, conflicting decisions, missing/foreign/duplicate IDs, changed narratives, invalid mechanism labels, unresolved recommendations, missing exclusion/resolution notes, stale candidate/policy identities and formula-based attestations. The new workflow tests reject a taxonomy gap without human resolution and an independent numerical discrepancy before publication. Invented-only end-to-end tests exercise review export, the guarded finalizer, provenance loading, precise arithmetic and preserved inputs; production validators expose no fixture-bypass CLI.

The focused Human review form preserves the original eight inclusion cases and nine mechanism mappings as its priority queue, while retaining all 188 records and the original audit/source sheets. Six additional invented-record tests verify that the reader uses explicit entries from this form and rejects ID/narrative mismatches, changed AI proposals, altered adjudication flags and conflicting legacy human inputs. Progress formulas report entries present, not approval. The form reader does not generate source values, reviewer identities or approvals. Ryan subsequently explicitly confirmed his personal review of the original 171 entries. The assistant recorded his new stated decisions separately in human correction fields, preserving all original AI/source cells; four conditional mechanism entries remain unapproved.

Release-path tests reject environments, raw downloads, nested datasets, candidate/review CSVs, workbooks, audit reports and secrets. The prepared directory now has a strict publication allowlist: only the 2025 runtime CSV/provenance, codebook and public preparation README. This closes a gap where a differently named policy or approval report could otherwise enter an archive. The read-only provenance audit flags assistant-transcribed attestations without changing them or treating populated Boolean cells as proof of review. A new Git regression stages explicitly invented CRLF files and verifies that the hashed CSV/provenance/codebook bytes survive

unchanged. Explicit -text attributes prevent a local-valid/cloud-invalid bundle caused by newline conversion. The committed-HEAD audit found no private paths or candidate secret patterns, but blocked release for missing reviewed runtime data, pending links and uncommitted work. This is a limited pattern/path audit, not proof that no secret exists anywhere in history. The final staged diff and HEAD require inspection; the ZIP guard refuses release while the runtime pair and public links are missing.

Dependency checks passed: all 45 installed packages compatible and the lockfile consistent with the declared manifest. Local Streamlit health and root responded HTTP 200; AppTest raised zero exceptions and displayed the expected unavailable-data message. This verifies honest missing-data behavior, not a working public real-data release. The version 1.1 migration updates preparation/transfer/loading/codebook and sidebar contracts. Filters/charts/exports remain data-driven and preserve the numerical definitions. Dependency pins and one-shot-prompt.md are unchanged. CR-07 version assertions now expect 1.1 while preserving its original behavior checks. The two already committed Colby property tests also passed: record-order invariance and unchanged shares/counts under uniform weight scaling.

PENDING RELEASE EVIDENCE: four actual mechanism decisions, independently checked real-data outputs, Linux/cloud behavior, public/incognito access, live screenshots, realistic user interpretation and the recorded demo. Automated success cannot establish truthful human review, public availability or practical decision value. The professor's 03-mvp.pdf requires those remaining items; this PDF stays an evidence draft until they are completed.